

Volatility and Value-at-Risk Forecasting using GARCH-type Models across Vietnamese Equity Size Segments: Evidence from VN30, VNMidcap, VNSmallcap and VNAllshare

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Abstract—This paper investigates one-day-ahead volatility and Value-at-Risk (VaR) forecasting across Vietnamese equity size segments. The study uses daily closing prices of VN30, VNMidcap, VNSmallcap and VNAllshare over the common sample from 20 November 2014 to 31 December 2025, and constructs daily log returns from 21 November 2014 to 31 December 2025. The empirical framework estimates AR(1)-GARCH-type models, including GARCH, GJR-GARCH, EGARCH and APARCH specifications, under Normal, Student- t and Skewed Student- t innovation distributions. Model adequacy is evaluated using residual diagnostics, while VaR performance is assessed for long and short trading positions at the 1% and 5% levels using in-sample and rolling out-of-sample backtesting. The results show that Vietnamese equity returns are stationary but non-normal, negatively skewed, fat-tailed and conditionally heteroskedastic. EGARCH with Skewed Student- t innovations provides the best in-sample fit for all four indices. However, the rolling out-of-sample VaR results indicate that GARCH(1,1) with Skewed Student- t innovations achieves the strongest overall backtesting performance. The evidence suggests that distributional asymmetry is more important than volatility-equation complexity for VaR forecasting in this sample.

Index Terms—Value-at-Risk, volatility forecasting, GARCH, GJR-GARCH, EGARCH, APARCH, Skewed Student- t distribution, Vietnamese stock market, equity size segments.

I. INTRODUCTION

Financial return series rarely behave as independent observations with constant variance. Equity returns typically exhibit volatility clustering, fat tails, skewness, and asymmetric responses to market news. These empirical properties matter directly for risk management because they affect the probability and severity of extreme losses. A model that treats returns as normally distributed with constant variance may describe average movements reasonably well, but it will usually misrepresent the tails of the distribution. This problem becomes especially important when the objective is Value-at-Risk (VaR) forecasting, because VaR is defined by tail quantiles rather than by the conditional mean of returns.

VaR has become one of the standard tools for measuring market risk because it summarises potential loss in a single interpretable number. Under the Basel market-risk framework, financial institutions were allowed to use internal models for capital adequacy calculations, which increased the practical importance of reliable VaR measurement [20]. However, the usefulness of VaR depends on the accuracy of the conditional distribution used to generate the tail forecast. If a model

underestimates tail probabilities, it produces a VaR measure that is too low and therefore provides inadequate capital protection. If a model is too conservative, it may overstate risk and impose unnecessary capital or margin costs. Thus, a useful VaR model must balance statistical accuracy and economic relevance.

GARCH-type models are widely used for VaR forecasting because they explicitly model time-varying conditional volatility. The standard GARCH model of Bollerslev [2] captures volatility persistence through lagged squared shocks and lagged conditional variances. Extensions such as GJR-GARCH [4], EGARCH [3], and APARCH [5] allow for asymmetric volatility responses, which are important when negative shocks affect future volatility differently from positive shocks of the same magnitude. In addition to the variance equation, the assumed innovation distribution is crucial. Normal innovations are often too thin-tailed for equity returns, while Student- t innovations allow fat tails but remain symmetric. Skewed Student- t innovations are more flexible because they can capture both excess kurtosis and left-right tail asymmetry. For VaR forecasting, this distinction is central because long-position VaR depends on the left tail of returns, whereas short-position VaR depends on the right tail.

This paper studies volatility and one-day-ahead VaR forecasting across Vietnamese equity size segments. Specifically, it examines four indices: VN30, VNMidcap, VNSmallcap, and VNAllshare. These indices represent large-cap, mid-cap, small-cap, and broad-market exposures within the Vietnamese equity market. The analysis uses daily closing prices from 20 November 2014 to 31 December 2025 and constructs daily log returns from 21 November 2014 to 31 December 2025. The common sample contains 2,776 return observations for each index. The focus on size segments is intentional, but it does not assume that these indices behave as independent markets. Since all four indices belong to the same national equity market, they are expected to share a strong common market component and to react to the same macro-financial shocks. The empirical question is therefore more specific: whether this common market component is strong enough to justify a single VaR specification for all segments, or whether size-specific differences in liquidity, investor composition, volatility persistence, skewness, and tail behaviour still require separate model selection and backtesting. In this sense, the paper studies whether a VaR model that works for VN30 can be transferred

mechanically to VNMidcap or VNSmallcap.

The Vietnamese market provides a useful setting for this question. Vietnam is an emerging equity market with strong retail participation, changing market depth, and visible differences in liquidity across firm-size groups. These features may amplify non-normality, volatility clustering, and tail asymmetry. At the same time, most VaR studies either focus on developed markets, compare broad indices across countries, or evaluate only downside risk for long positions. Less attention has been paid to whether VaR model performance differs across equity size segments within one market. This gap matters because risk management is not conducted only at the aggregate market level. Portfolio managers, proprietary traders, and institutional investors often hold exposures tilted toward particular size groups. A single broad-market risk model may therefore be misleading if the tail properties of the segments are materially different.

The closest methodological foundation for this paper is the long-short VaR framework of Giot and Laurent [12] and Diamandis et al. [13]. Giot and Laurent show that skewed density models are important when both left and right return tails are evaluated. Diamandis et al. extend this framework to developed and emerging equity markets and find that Skewed Student- t APARCH models improve one-day-ahead VaR forecasts for both long and short trading positions. This paper builds on that logic but changes the empirical design. Instead of comparing many countries, it compares size segments within Vietnam. This allows the study to ask whether the international finding that skewed fat-tailed models improve VaR forecasting also holds inside a single emerging market, and whether the most flexible volatility equation remains superior under rolling out-of-sample evaluation.

The paper addresses four research questions. First, do the daily returns of VN30, VNMidcap, VNSmallcap, and VNAI-share exhibit statistical properties that justify GARCH-type modelling? Second, which GARCH-type specification provides the best in-sample fit and residual diagnostic performance for each size segment? Third, which model performs best in one-day-ahead VaR forecasting under in-sample and rolling out-of-sample backtesting? Fourth, does VaR performance differ between long and short positions, between the 1% and 5% VaR levels, and across equity size segments?

These questions lead to four hypotheses. The first hypothesis is that Vietnamese equity returns are stationary but non-normal, fat-tailed, negatively skewed, and conditionally heteroskedastic. The second hypothesis is that Student- t or Skewed Student- t innovations outperform Normal innovations in VaR forecasting, with the Skewed Student- t distribution expected to perform best because it captures both fat tails and asymmetry. The third hypothesis is that asymmetric variance equations such as GJR-GARCH, EGARCH, and APARCH improve volatility modelling relative to symmetric GARCH(1,1). The fourth hypothesis is that VaR accuracy and risk levels differ across size segments and between long and short positions, reflecting differences in tail behaviour and volatility dynamics.

To test these hypotheses, the paper estimates AR(1)-GARCH-

type models for each index. The variance specifications include GARCH(1,1), GJR-GARCH(1,1), EGARCH(1,1), and APARCH(1,1). Each specification is estimated under Normal, Student- t , and Skewed Student- t innovations where applicable, producing a set of competing model-distribution combinations. Model adequacy is first evaluated using residual diagnostics, including Ljung-Box tests on standardised residuals, Ljung-Box tests on squared standardised residuals, and ARCH-LM tests. VaR performance is then evaluated for long and short positions at the 1% and 5% levels. The backtesting framework includes the Kupiec unconditional coverage test [8], the Christoffersen conditional coverage test [9], the Dynamic Quantile test [10], and magnitude-based measures such as Expected Shortfall and the Average Multiple of Tail Event to VaR.

The paper deliberately separates in-sample fit from forecasting performance. In-sample likelihood criteria such as AIC and BIC are useful for identifying models that describe the observed conditional distribution well. However, VaR is fundamentally a forecasting problem. A model is useful only if it generates accurate tail forecasts on observations not used in estimation. For this reason, the main evaluation is based on a rolling out-of-sample backtesting scheme. The model is initially estimated using a 1,000-day rolling window and then used to produce one-day-ahead VaR forecasts. Parameters are updated periodically as the window advances through the sample. This design gives a more demanding test of model performance than full-sample in-sample evaluation.

The empirical findings are clear. Preliminary tests show that all four Vietnamese return series are stationary, non-normal, fat-tailed, negatively skewed, and conditionally heteroskedastic. These facts support the use of GARCH-type volatility models with non-Gaussian innovations. In the in-sample estimation, EGARCH with Skewed Student- t innovations achieves the lowest AIC and BIC for all four indices. This indicates that asymmetric volatility dynamics and skewed fat-tailed innovations improve in-sample density fit. However, the rolling out-of-sample VaR results lead to a different ranking: GARCH(1,1) with Skewed Student- t innovations delivers the strongest overall backtesting performance, with a 75.0% main pass rate across the index-position-level combinations. Thus, the model with the best in-sample fit is not necessarily the best VaR forecasting model.

The results also show that distributional specification matters more than variance-equation complexity. All best out-of-sample models use the Skewed Student- t distribution, while models with symmetric Student- t innovations perform poorly in rolling VaR backtesting. This means that modelling both fat tails and skewness is essential for Vietnamese equity VaR forecasting. The long-short comparison further shows that downside risk is harder to forecast than upside risk: long-position failure rates exceed short-position failure rates at the 1% level for all four indices. Finally, the size-segment comparison reveals that VNMidcap has the highest average absolute VaR, while VNSmallcap has the strongest estimated left skewness and the weakest residual diagnostic performance. Hence, risk

differs across size segments not only in magnitude but also in distributional shape and model adequacy.

The contribution of this paper is fourfold. First, it applies a long-short VaR framework to Vietnamese equity size segments rather than to broad international market indices. This allows the paper to examine whether size-segment differences remain relevant even when the indices share a common national market factor. Second, it compares several GARCH-type specifications under multiple innovation distributions, allowing the study to separate the effect of variance-equation choice from the effect of distributional choice. Third, it evaluates both statistical coverage and economic severity using backtesting tests, Expected Shortfall, and tail-event multiples. Fourth, it shows that for Vietnamese equity indices, the Skewed Student- t distribution is more important than selecting the most complex variance equation. This finding is directly relevant for practical risk management: a parsimonious GARCH(1,1)-SkewT model may provide more reliable rolling VaR forecasts than more flexible models that fit better in-sample.

The remainder of the paper is organised as follows. Section II reviews the literature on VaR, GARCH-type models, conditional distributions, and long-short position VaR. Section III describes the data, model specifications, VaR calculation, and backtesting framework. Section IV reports the empirical results, including preliminary tests, model estimation, residual diagnostics, and in-sample and out-of-sample VaR backtesting. Section V discusses the economic and methodological implications, limitations, and possible extensions. Section VI concludes.

II. LITERATURE REVIEW

A. VaR, GARCH-Type Models, and Conditional Distributions

Value-at-Risk (VaR) is one of the most widely used measures of market risk. It summarises the maximum expected loss over a given horizon at a specified confidence level. For instance, a one-day 1% VaR gives the loss threshold that should be exceeded only 1% of the time if the model is correctly specified. Its practical appeal lies in reducing market risk to a single interpretable number that can be used for internal risk control, capital allocation, and regulatory monitoring. VaR became especially important after the Basel market-risk framework allowed financial institutions to use internal risk models for capital adequacy purposes [20]. Standard references such as Jorion [6] and Alexander [7] emphasise that the reliability of VaR depends critically on the accuracy of the conditional return distribution used to compute tail quantiles.

The main statistical challenge is that financial returns rarely behave as independent Gaussian observations with constant variance. Equity returns usually exhibit volatility clustering, excess kurtosis, skewness, and asymmetric responses to good and bad news. The ARCH model introduced by Engle [1] was the first major framework for modelling time-varying conditional variance. Bollerslev [2] generalised this approach into the GARCH model, where conditional volatility depends on both lagged squared shocks and lagged conditional variances. The GARCH(1,1) specification remains a standard benchmark

because it is parsimonious and captures volatility persistence effectively.

However, standard GARCH combined with normally distributed innovations has two limitations for VaR forecasting. First, the Normal distribution tends to underestimate tail probabilities because daily stock returns are fat-tailed. Second, the symmetric GARCH variance equation does not distinguish between the volatility impact of positive and negative shocks. This is restrictive because equity markets often exhibit a leverage effect, where negative shocks raise future volatility more strongly than positive shocks of the same magnitude. Several GARCH extensions address this issue. Nelson's EGARCH model specifies the logarithm of conditional variance and allows volatility to respond asymmetrically to standardised shocks [3]. The GJR-GARCH model adds an indicator term for negative shocks and directly captures the additional effect of bad news on volatility [4]. The APARCH model of Ding, Granger, and Engle [5] is even more flexible because it introduces both asymmetric effects and a power transformation of volatility.

Distributional assumptions are equally important for VaR. Bollerslev's Student- t conditional distribution improves on the Normal distribution by allowing heavier tails [16]. Nevertheless, the standard Student- t distribution remains symmetric. It can model extreme observations in both tails, but it cannot represent the empirical possibility that the left and right tails have different shapes. Fernandez and Steel [17] introduced a skewed Student- t formulation that allows both fat tails and asymmetry, while Lambert and Laurent [18] adapted the skewed Student- t density for GARCH-type volatility models. This extension is particularly relevant for VaR because long-position risk depends on the left tail of returns, whereas short-position risk depends on the right tail.

Backtesting is the final component of VaR modelling. A VaR model is not adequate merely because it fits the data in-sample; it must also generate violations at the correct frequency and without predictable clustering. Kupiec [8] proposed the unconditional coverage test, which checks whether the empirical violation rate equals the nominal VaR level. Christoffersen [9] extended this logic by testing both coverage and independence of violations. Engle and Manganelli [10] proposed the Dynamic Quantile (DQ) test, which evaluates whether VaR violations are predictable from past information. These tests are central because VaR is ultimately a forecasting and risk-control problem, not only an in-sample density-fitting exercise.

B. Empirical Evidence on Long and Short Position VaR

A large part of the VaR literature focuses on long-position risk, where losses occur when asset prices fall. In that case, VaR is computed from the left tail of the return distribution. However, traders may also hold short positions, where losses occur when asset prices rise. Short-position VaR therefore depends on the right tail of the return distribution. If returns are asymmetric, the two tails cannot be treated as mirror images. This distinction motivates the joint evaluation of long and short VaR.

Giot and Laurent [12] provide one of the most important studies in this area. They evaluate VaR for both long and short trading positions using daily data for major equity indices, including FTSE100, NASDAQ, and NIKKEI. Their main result is that models with skewed Student- t innovations, especially Skewed Student- t APARCH, perform better than models based on Normal or symmetric Student- t innovations. The reason is direct: when both tails of the return distribution matter, a symmetric conditional density is too restrictive.

Diamandis et al. [13] extend the long-short VaR framework to a broader international setting. They examine 21 developed and emerging stock markets and compare RiskMetrics, Normal APARCH, Student- t APARCH, and Skewed Student- t APARCH models. Their evidence shows that Skewed Student- t APARCH improves one-day-ahead VaR forecasts for both long and short trading positions. This study is the closest methodological benchmark for the present paper because it combines three elements that are also central here: GARCH-type volatility modelling, skewed fat-tailed innovations, and separate evaluation of long and short VaR.

Other empirical studies support the broader conclusion that asymmetric volatility and flexible conditional distributions improve VaR forecasts. So and Yu [21] compare alternative GARCH-type VaR models and find that specifications allowing for asymmetric effects and non-Gaussian innovations often outperform simpler alternatives. McMillan and Speight [22] provide additional evidence that asymmetric and long-memory volatility models are relevant for market-risk forecasting in emerging equity markets. At the same time, Hansen and Lunde [23] show that the simple GARCH(1,1) model is difficult to outperform in out-of-sample volatility forecasting. This point is important because a more flexible volatility equation may improve in-sample fit without necessarily improving out-of-sample VaR accuracy.

Evidence from Southeast Asian markets also supports the use of non-normal VaR models. Emerging and frontier-style equity markets often exhibit stronger volatility clustering, sharper drawdowns, lower liquidity, and more pronounced non-normality than developed markets. Bui Quang et al. [14] compare VaR methods for Southeast Asian stock markets using GARCH-type models, filtered historical simulation, and stochastic volatility models. Their results show that VaR performance varies across countries and model classes, and that simple assumptions are often insufficient for reliable risk forecasting. This evidence makes Vietnam a relevant setting for testing whether the international findings on skewed fat-tailed GARCH-type VaR models carry over to an emerging equity market.

C. Research Gap and Positioning of This Study

Although the existing literature provides strong evidence in favour of GARCH-type VaR models with flexible conditional distributions, several gaps remain.

First, most previous studies compare VaR models across countries or broad market indices. Much less attention has been paid to VaR forecasting across equity size segments within a

single market. This is not a trivial extension: within one national market, size indices may share a strong common component, but still differ in tail asymmetry, volatility persistence, liquidity exposure, and model adequacy. The relevant question is therefore whether common market co-movement eliminates the need for segment-specific VaR modelling, or whether tail-risk differences remain large enough to justify separate models. This distinction matters because large-cap, mid-cap, and small-cap stocks differ in liquidity, information flow, investor base, trading activity, and sensitivity to market stress. A model that performs well for a large-cap benchmark may not be appropriate for a mid-cap or small-cap index.

Second, many studies on Vietnam and Southeast Asia emphasise downside risk for long investors. This is useful but incomplete. A full VaR framework should evaluate both long and short positions because these two positions correspond to different tails of the return distribution. If returns are skewed, then left-tail and right-tail VaR performance may differ substantially. Ignoring short-position VaR prevents a direct assessment of whether the model captures the full conditional return distribution.

Third, the framework of Giot and Laurent [12] and Diamandis et al. [13] has not been extensively examined across Vietnamese equity size segments. Vietnam is a relevant empirical setting because it combines emerging-market characteristics with strong liquidity differences across firm-size groups and market microstructure features that may affect measured volatility. It is therefore not obvious that the Skewed Student- t APARCH dominance documented in international markets will hold unchanged in Vietnam. In particular, a more flexible variance equation may not necessarily dominate a simpler GARCH model once performance is evaluated with rolling out-of-sample backtesting.

This paper addresses these gaps by applying the long-short VaR framework of Giot and Laurent [12] and Diamandis et al. [13] to four Vietnamese equity size segments: VN30, VN-Midcap, VNSmallcap, and VNAllshare. Instead of comparing countries, the paper compares risk dynamics within one market across large-cap, mid-cap, small-cap, and broad-market indices. This design isolates size-segment differences while holding the national market environment fixed.

The study contributes in three ways. First, it evaluates whether GARCH-type models with Normal, Student- t , and Skewed Student- t innovations can adequately model Vietnamese equity returns. Second, it computes VaR for both long and short trading positions at the 1% and 5% levels, thereby examining both the left and right tails of the return distribution. Third, it compares in-sample model fit with rolling out-of-sample VaR backtesting. This final step is essential because the best-fitting model is not necessarily the best forecasting model.

The paper therefore sits at the intersection of three strands of literature: GARCH-type volatility modelling, long-short VaR backtesting, and emerging market risk measurement. Its central empirical question is whether the international conclusion that skewed fat-tailed distributions improve VaR forecasting

continues to hold across Vietnamese equity size segments, and whether the most flexible volatility equation remains superior once forecasting performance is evaluated out of sample.

III. DATA AND METHODOLOGY

A. Data

The empirical analysis uses daily closing prices for four Vietnamese equity indices: VN30, VNMidcap, VNSmallcap and VNAllshare. Raw data are collected from Investing.com at daily frequency. After cleaning and aligning all four series, the common closing-price sample runs from 20 November 2014 to 31 December 2025. Daily log returns are then computed from 21 November 2014 to 31 December 2025, yielding 2,776 return observations per index.

The four indices represent distinct equity size segments of the Vietnamese market. VN30 tracks the 30 largest and most liquid stocks listed on the Ho Chi Minh Stock Exchange (HOSE), and serves as the blue-chip benchmark. VNMidcap covers medium-capitalisation stocks on HOSE, while VNSmallcap covers small-capitalisation stocks on the same exchange. VNAllshare is a broad market index that aggregates all eligible HOSE listings across all capitalisation tiers. Using all four indices allows a direct comparison of volatility dynamics and VaR accuracy across size segments within a single, unified framework. Since the indices belong to the same market, they are expected to share common shocks; the empirical focus is therefore on whether tail-risk and forecasting performance differ after accounting for this shared market environment.

The sample period is deliberately chosen to include several episodes of elevated market stress, notably the COVID-19 shock of early 2020 and the broad equity sell-off of 2022. Including these periods is important because VaR models are designed precisely to quantify tail risk; evaluating them only on calm data would give an incomplete picture of their practical reliability.

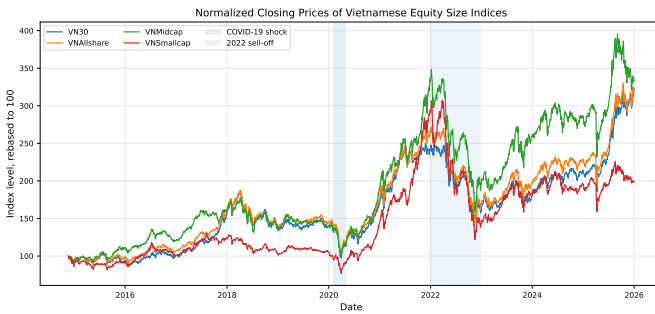


Fig. 1. Normalized closing prices of VN30, VNAllshare, VNMidcap and VNSmallcap. All indices are rebased to 100 at the beginning of the common sample.

Figure 1 shows that the four indices share a strong common market component. The broad phases of expansion, stress, and recovery are visible across all size segments, especially around the COVID-19 shock and the 2022 equity sell-off. This common movement is expected within a single national

equity market. The purpose of the analysis is therefore not to claim that the indices are independent, but to test whether their tail-risk and VaR-forecasting properties still differ despite this common component.

The daily log return is defined as

$$r_t = 100 \times \ln\left(\frac{P_t}{P_{t-1}}\right), \quad (1)$$

where P_t is the closing price on day t and returns are multiplied by 100 so that all series are measured in percentage points.

The data-processing procedure follows five steps. First, raw Vietnamese column names are standardised into English variable names. Second, trading dates are parsed and sorted chronologically. Third, all price fields are cleaned and converted to numeric format. Fourth, invalid observations and alignment inconsistencies are identified and removed. Fifth, the four indices are merged into a common sample and daily log returns are computed using (1).

TABLE I
DESCRIPTIVE STATISTICS OF DAILY LOG RETURNS

Statistic	VN30	VNMidcap	VNSmallcap	VNAllshare
Observations	2,776	2,776	2,776	2,776
Mean	0.042	0.043	0.025	0.042
Std. deviation	1.221	1.275	1.143	1.184
Skewness	-0.816	-1.060	-1.231	-0.986
Excess kurtosis	4.784	4.746	5.492	5.335
Minimum	-7.052	-6.898	-6.616	-6.987
Maximum	6.670	6.472	6.433	6.607
$LB^2(10)$	691.0***	811.4***	1050.3***	713.9***
JB	2,943.2***	3,113.3***	4,174.8***	3,726.9***
ADF	-50.43***	-12.42***	-12.25***	-49.52***

Notes: Returns are measured in percentage points. $LB^2(10)$ is the Ljung–Box statistic on squared returns with 10 lags. JB is the Jarque–Bera normality test statistic. ADF is the augmented Dickey–Fuller test statistic. *** denotes significance at the 1% level.

Table I reports descriptive statistics and preliminary diagnostics for the four daily return series. All indices display standard stylised facts of equity returns: daily means are close to zero, skewness is negative, and excess kurtosis is positive, indicating asymmetric and fat-tailed return distributions. The Ljung–Box statistics on squared returns are highly significant for all indices, confirming volatility clustering. Jarque–Bera tests reject normality for all return series at the 1% level, while augmented Dickey–Fuller tests reject the unit-root null and therefore support stationarity. These results justify the use of GARCH-type models with non-normal innovation distributions.

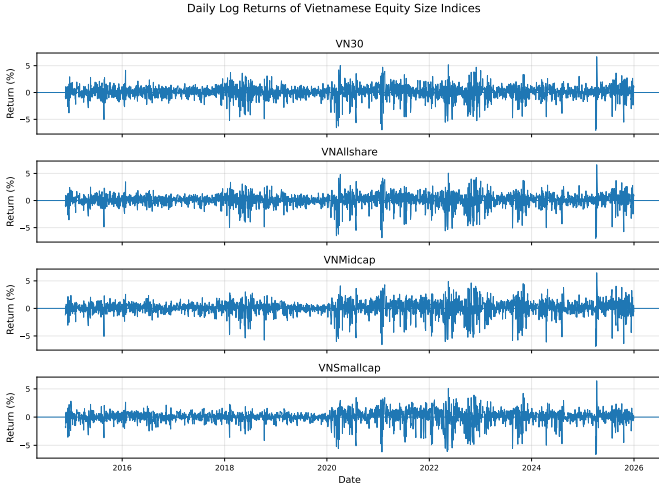


Fig. 2. Daily log returns of the four Vietnamese equity indices. The plots show clustered volatility and large return shocks during market stress episodes.

Figure 2 visually supports the diagnostic evidence in Table I. Large absolute returns are not evenly distributed over time; instead, they cluster during turbulent market periods. The return plots also show that shocks often occur at similar dates across indices, again indicating a common market component. However, the amplitude and persistence of these shocks are not identical across segments, which motivates the later comparison of conditional volatility and VaR performance.

B. GARCH-Type Models

1) *Mean Equation*: All models share a common AR(1) mean equation,

$$r_t = \mu + \phi_1 r_{t-1} + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad (2)$$

where μ is the intercept, ϕ_1 is the first-order autoregressive coefficient, $\sigma_t > 0$ is the conditional standard deviation, and z_t is an i.i.d. innovation with zero mean and unit variance. The AR(1) specification is included to remove any residual serial correlation in returns, which would otherwise contaminate the heteroskedasticity tests; it is the most parsimonious correction consistent with the Ljung–Box diagnostics reported in Table I.

2) *Variance Equations*: Four variance specifications are estimated.

GARCH(1,1). The standard GARCH(1,1) model of Bollerslev [2] specifies

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad (3)$$

with $\omega > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$ for covariance stationarity.

GJR-GARCH(1,1). The GJR model of Glosten, Jagannathan and Runkle [4] augments (3) with an asymmetry term:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \gamma_1 \mathbf{1}_{[\varepsilon_{t-1} < 0]} \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \quad (4)$$

A positive γ_1 implies that negative shocks increase conditional variance by more than positive shocks of equal magnitude, consistent with the leverage effect documented in equity markets.

EGARCH(1,1). Nelson’s [3] exponential GARCH model specifies the log-variance as

$$\ln \sigma_t^2 = \omega + \alpha_1 (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma_1 z_{t-1} + \beta_1 \ln \sigma_{t-1}^2. \quad (5)$$

The parameter $\gamma_1 < 0$ captures the leverage effect without imposing non-negativity constraints on the coefficients, and the log-variance formulation guarantees $\sigma_t^2 > 0$ by construction.

APARCH(1,1). The asymmetric power ARCH model of Ding, Granger and Engle [5] generalises (3) via a Box–Cox transformation of σ_t :

$$\sigma_t^\delta = \omega + \alpha_1 (|\varepsilon_{t-1}| - \gamma_1 \varepsilon_{t-1})^\delta + \beta_1 \sigma_{t-1}^\delta, \quad (6)$$

where $\delta > 0$ is the power parameter and $\gamma_1 \in (-1, 1)$ governs sign asymmetry. When $\delta = 2$ and $\gamma_1 = 0$, (6) reduces to the standard GARCH(1,1).

C. Innovation Distributions

Three distributional assumptions for z_t are considered.

Normal. $z_t \sim \mathcal{N}(0, 1)$ is the baseline. It is well known to underestimate tail probabilities for daily returns and is included chiefly as a benchmark.

Student- t . $z_t \sim t(0, 1, \nu)$, where $\nu > 2$ is the degrees-of-freedom parameter estimated jointly with the variance parameters [16]. The heavier tails of the Student- t improve VaR coverage relative to the Normal, but the symmetric distribution cannot accommodate the negative skewness evident in Table I.

Skewed Student- t . Following Fernandez and Steel [17] as re-parameterised by Lambert and Laurent [18], the standardised skewed Student- t density is

$$f(z | \xi, \nu) = \begin{cases} \frac{2}{\xi + \xi^{-1}} g(\xi s z + m | \nu) & z < -m/s, \\ \frac{2}{\xi + \xi^{-1}} g(s z / \xi + m | \nu) & z \geq -m/s, \end{cases} \quad (7)$$

where $g(\cdot | \nu)$ is the symmetric unit-variance Student- t density, $\xi > 0$ is the asymmetry parameter, and m and s^2 are the mean and variance of the non-standardised skewed Student- t that ensure z_t has zero mean and unit variance. $\log(\xi) < 0$ indicates left skewness and $\log(\xi) > 0$ right skewness. All nine model–distribution combinations are estimated by maximum likelihood using the `arch` library in Python.

D. Value-at-Risk Computation

Following Giot and Laurent [12], VaR is defined as the maximum loss on a portfolio over a one-day horizon at a specified confidence level. For a trader holding a *long position*, the risk is that the asset price falls; for a *short position*, the risk is that it rises. This distinction requires modelling both tails of the return distribution.

Given the fitted conditional mean μ_t and conditional standard deviation σ_t , the one-day-ahead VaR forecasts are

$$\widehat{\text{VaR}}_t^{\text{long}}(\alpha) = \mu_t + q_\alpha^D \sigma_t, \quad (8)$$

$$\widehat{\text{VaR}}_t^{\text{short}}(\alpha) = \mu_t + q_{1-\alpha}^D \sigma_t, \quad (9)$$

where q_α^D and $q_{1-\alpha}^D$ are the α -th and $(1-\alpha)$ -th quantiles of the chosen distribution D . Long VaR models the left tail (negative returns exceed the threshold), while short VaR models the right tail. Two significance levels are considered: $\alpha \in \{1\%, 5\%\}$.

E. Backtesting Framework

Model performance is evaluated through both in-sample and out-of-sample backtesting.

1) *In-Sample Backtesting*: In-sample VaR forecasts are computed as one-step-ahead predictions evaluated over the full estimation sample. This is equivalent to checking the internal consistency of each model on the data used to fit it, and it closely mirrors the approach of Giot and Laurent [12] and Diamandis et al. [13].

2) *Rolling Out-of-Sample Backtesting*: Out-of-sample evaluation uses a rolling-window scheme. The model is first estimated on an initial window of 1,000 trading days (approximately four years). The estimated parameters are then used to produce a one-step-ahead VaR forecast for the next trading day. The window advances by one day and the model is re-estimated every 20 trading days to update the parameter estimates while limiting computational cost. This procedure is repeated until the end of the sample, generating a sequence of out-of-sample VaR forecasts for each model–distribution combination and each index. Out-of-sample evaluation is considered the more stringent test because it assesses genuinely ex-ante forecasting performance on data unseen during estimation.

3) *Backtesting Tests*: Let $\mathbf{1}_t = \mathbf{1}\{r_t < -\widehat{\text{VaR}}_t^{\text{long}}\}$ denote the *violation indicator* for the long position (and analogously for the short position). A correctly specified VaR model requires (i) that violations occur at the right frequency and (ii) that violations are not serially correlated.

Kupiec [8] likelihood ratio test. The unconditional coverage test examines whether the empirical failure rate $\hat{f} = N/T$, where N is the number of violations in T observations, equals the nominal level α :

$$\text{LR}_{\text{uc}} = 2 \left[\ln(\hat{f}^N (1 - \hat{f})^{T-N}) - \ln(\alpha^N (1 - \alpha)^{T-N}) \right] \xrightarrow{d} \chi_1^2. \quad (10)$$

Christoffersen [9] conditional coverage test. This test jointly checks unconditional coverage and independence of violations by decomposing the likelihood ratio into two components: $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}}$, where LR_{ind} tests the null hypothesis that $\mathbf{1}_t$ is independent of $\mathbf{1}_{t-1}$. The statistic is asymptotically χ_2^2 .

Dynamic Quantile (DQ) test [10]. Define the centred hit sequence $H_t = \mathbf{1}_t - \alpha$. The DQ test regresses H_t on five of its own lags and on the current VaR forecast, and tests the null that all slope coefficients are zero:

$$\text{DQ} = \frac{\hat{\beta}' \mathbf{X}' \mathbf{X} \hat{\beta}}{\alpha(1 - \alpha)} \xrightarrow{d} \chi_6^2. \quad (11)$$

A significant DQ statistic indicates that current violations are predictable from past information, which constitutes a more refined failure than the Kupiec test alone can detect.

4) *Magnitude Measures*: Two additional statistics quantify the severity of violations rather than their frequency.

The *Expected Shortfall* (ES) is the average loss conditional on a violation occurring:

$$\widehat{\text{ES}}(\alpha) = \frac{1}{N} \sum_{t: r_t < -\widehat{\text{VaR}}_t} (-r_t). \quad (12)$$

The *Average Multiple of Tail Event to Risk Measure* (AM) [19] normalises each violation by the corresponding VaR forecast:

$$\widehat{\text{AM}}(\alpha) = \frac{1}{N} \sum_{t: r_t < -\widehat{\text{VaR}}_t} \frac{-r_t}{\widehat{\text{VaR}}_t}. \quad (13)$$

A value of $\widehat{\text{AM}} > 1$ means that, on average, actual losses exceeded the VaR forecast; a value close to 1 indicates that the model underestimates tail losses only marginally when it fails. Together, ES and AM complement the pass/fail verdict of the Kupiec and Christoffersen tests by revealing how conservative or reckless a model is when its coverage breaks down.

IV. EMPIRICAL RESULTS

A. Preliminary Time-Series Tests

Before estimating the GARCH-type models, all four return series are subjected to preliminary tests for normality, stationarity, serial correlation, and conditional heteroskedasticity. The purpose is to verify whether the modelling choices described in Section III are empirically justified.

Table II reports the test statistics and corresponding p -values. The Jarque–Bera test rejects the null hypothesis of normality for all four series at the 1% level. The Augmented Dickey–Fuller test rejects the unit-root null for every return series, confirming stationarity. The Ljung–Box test on raw returns does not reject no autocorrelation for VN30 at the 5% level, but rejects it for VNAllshare, VNMidcap, and VNSmallcap. The evidence is especially strong for VNMidcap and VNSmallcap, supporting the use of an AR(1) conditional mean equation. More importantly, the Ljung–Box test on squared returns and the ARCH-LM test strongly reject the absence of second-moment dependence for all indices. Hence, conditional heteroskedasticity is a systematic feature of the data.

TABLE II
PRELIMINARY TIME-SERIES TESTS

Test	VN30	VNAllshare	VNMidcap	VNSmallcap
<i>JB</i> statistic	2943.179	3726.894	3113.294	4174.791
<i>JB</i> <i>p</i> -value	<0.001	<0.001	<0.001	<0.001
<i>ADF</i> statistic	-50.426	-49.523	-12.418	-12.255
<i>ADF</i> <i>p</i> -value	<0.001	<0.001	<0.001	<0.001
<i>LB</i> (10) statistic	16.796	20.090	44.235	122.806
<i>LB</i> (10) <i>p</i> -value	0.079	0.028	<0.001	<0.001
<i>LB</i> ² (10) statistic	691.014	713.922	811.418	1050.255
<i>LB</i> ² (10) <i>p</i> -value	<0.001	<0.001	<0.001	<0.001
<i>ARCH-LM</i> (10) stat.	314.916	320.912	359.820	425.059
<i>ARCH-LM</i> (10) <i>p</i> -value	<0.001	<0.001	<0.001	<0.001

JB: Jarque–Bera normality test. *ADF*: Augmented Dickey–Fuller unit-root test. *LB*(10): Ljung–Box test on raw returns, 10 lags. *LB*²(10): Ljung–Box test on squared returns, 10 lags. *ARCH-LM*(10): Engle ARCH-LM test, 10 lags.

Taken together, the preliminary tests show that the Vietnamese equity return series are stationary but non-normal and conditionally heteroskedastic. These findings justify the use of GARCH-type models and non-Gaussian innovation distributions.

B. Correlation across Vietnamese Equity Size Segments

Before modelling each index independently, it is useful to examine the contemporaneous co-movement structure across the four equity size segments. Table III reports the static correlation matrix of daily log returns over the common sample.

TABLE III
CORRELATION MATRIX OF DAILY LOG RETURNS

	VN30	VNAllshare	VNMidcap	VNSmallcap
VN30	1.000	0.984	0.832	0.745
VNAllshare	0.984	1.000	0.908	0.828
VNMidcap	0.832	0.908	1.000	0.903
VNSmallcap	0.745	0.828	0.903	1.000

VN30 and VNAllshare have the highest correlation, which is expected because large-cap stocks dominate the broad market index. VNMidcap and VNSmallcap are also strongly correlated, reflecting their shared exposure to smaller-size market segments. The correlation between VN30 and VNSmallcap is the lowest among the reported pairs.

These correlations are consistent with Figures 1–3: the four indices clearly share a common market component. However, high correlation does not imply identical VaR dynamics. Correlation measures average co-movement, whereas VaR performance depends on tail behaviour, skewness, volatility persistence, and violation clustering. The correlation evidence therefore does not weaken the size-segment design; instead, it clarifies the empirical question. The issue is whether common market exposure is sufficient to make one risk model transferable across all segments. The remaining results show that it is not.

C. In-Sample Model Estimation Results

Table IV reports selected parameter estimates for the Skewed Student-*t* specifications, which are central to the analysis.

All 36 estimated models across the nine model–distribution combinations and four indices converged successfully.

TABLE IV
SELECTED PARAMETER ESTIMATES: SKEWED STUDENT-*t* MODELS

Index	Model	$\hat{\alpha}_1$	$\hat{\gamma}_1$	$\hat{\beta}_1$	Persist.	$\hat{\lambda}$
VN30	GARCH	0.126	—	0.837	0.963	-0.137
VN30	GJR	0.059	0.135	0.812	0.939	-0.137
VN30	EGARCH	0.252	-0.095	0.940	—	-0.134
VN30	APARCH	0.136	0.370	0.823	—	-0.135
VNAllshare	GARCH	0.138	—	0.829	0.967	-0.182
VNAllshare	GJR	0.056	0.167	0.797	0.936	-0.178
VNAllshare	EGARCH	0.269	-0.104	0.938	—	-0.179
VNAllshare	APARCH	0.152	0.421	0.809	—	-0.176
VNMidcap	GARCH	0.160	—	0.827	0.987	-0.212
VNMidcap	GJR	0.105	0.123	0.804	0.971	-0.205
VNMidcap	EGARCH	0.304	-0.074	0.952	—	-0.217
VNMidcap	APARCH	0.176	0.263	0.818	—	-0.211
VNSmallcap	GARCH	0.190	—	0.800	0.990	-0.247
VNSmallcap	GJR	0.154	0.071	0.792	0.982	-0.242
VNSmallcap	EGARCH	0.315	-0.047	0.958	—	-0.257
VNSmallcap	APARCH	0.182	0.171	0.818	—	-0.249

Persist. = $\hat{\alpha}_1 + \hat{\beta}_1$ for GARCH and $\hat{\alpha}_1 + 0.5\hat{\gamma}_1 + \hat{\beta}_1$ for GJR-GARCH. Persistence is not directly comparable for EGARCH and APARCH. $\hat{\lambda}$ is the skewness parameter of the Skewed Student-*t* distribution; $\hat{\lambda} < 0$ indicates left skewness. Most key volatility and distributional parameters are statistically significant at the 5% level, with exceptions mainly in the mean equation and selected leverage terms.

Three findings are important.

First, negative skewness is pervasive and size-dependent. The estimated $\hat{\lambda}$ is negative for all 16 model–index combinations, showing that left-tail risk is systematically present across all Vietnamese equity segments. Its magnitude becomes larger as market capitalisation falls, from approximately -0.135 for VN30 to about -0.25 for VNSmallcap. Economically, this means smaller-size segments have more asymmetric downside risk.

Second, leverage effects are broadly present. In the GJR-GARCH models, $\hat{\gamma}_1$ is positive for all indices, indicating that negative shocks raise volatility more than positive shocks of the same magnitude. However, the GJR leverage coefficient is weaker for VNSmallcap and should not be overstated. In the EGARCH specifications, $\hat{\gamma}_1 < 0$ across all indices; under Nelson’s parameterisation, this conveys the same economic interpretation that bad news has a larger effect on volatility than good news [3].

Third, volatility clustering is highly persistent. For the symmetric GARCH(1,1)-SkewT models, persistence rises from 0.963 for VN30 to 0.990 for VNSmallcap. Interpreted through a standard GARCH half-life calculation, this implies that volatility shocks decay much more slowly in smaller-size segments than in the large-cap segment.

Model selection by AIC and BIC. Across all four indices, EGARCH with Skewed Student-*t* innovations achieves the lowest AIC: 8012.0 for VN30, 7654.7 for VNAllshare, 7940.8 for VNMidcap, and 7108.2 for VNSmallcap. BIC rankings also select EGARCH-SkewT in every case. Therefore, the in-sample likelihood evidence supports asymmetric volatility dynamics and skewed fat-tailed innovations.

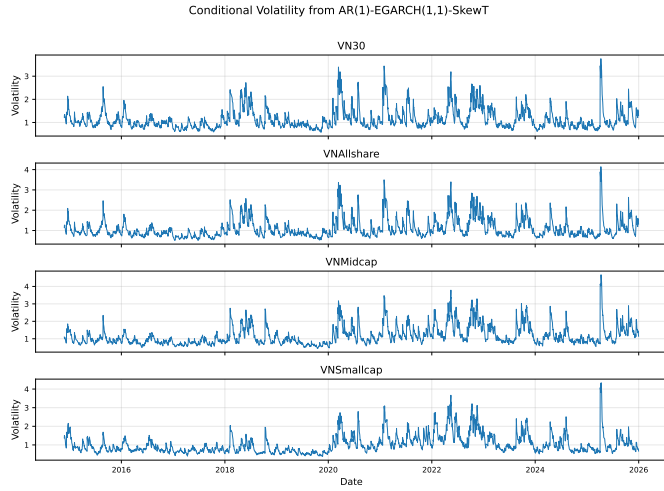


Fig. 3. Estimated conditional volatility from the EGARCH-SkewT model for each index. Volatility spikes are concentrated around major market stress periods.

Figure 3 translates the parameter estimates in Table IV into time-series volatility dynamics. Conditional volatility spikes occur in broadly similar periods across the four indices, confirming that they are exposed to common market-wide shocks. At the same time, the height and persistence of the volatility spikes differ across segments. This visual evidence is consistent with the parameter estimates: volatility clustering is common to all indices, but the strength and duration of volatility responses are segment-specific.

D. Residual Diagnostic Results

Table V summarises the residual diagnostics for all nine model specifications across the four indices. A model is marked as passing if it satisfies all three criteria at the 5% level: no autocorrelation in standardised residuals, no autocorrelation in squared standardised residuals, and no remaining ARCH effects.

TABLE V
RESIDUAL DIAGNOSTIC PASS/FAIL SUMMARY (LAG 10, 5% LEVEL)

Model	VN30	VNAllshare	VNMidcap	VNSmallcap
GARCH-Normal	✓	✓	✓	×
GARCH- t	✓	✓	×	×
GARCH-SkewT	✓	✓	×	×
GJR- t	✓	✓	✓	×
GJR-SkewT	✓	✓	×	×
EGARCH- t	✓	✓	✓	×
EGARCH-SkewT	✓	✓	×	×
APARCH- t	✓	✓	✓	×
APARCH-SkewT	✓	✓	×	×
Pass rate	9/9	9/9	4/9	0/9

✓: all three diagnostic criteria passed at the 5% level. ×: at least one criterion failed. The failed criterion is always the Ljung-Box test on standardised residuals; all models pass the squared-residual and ARCH-LM tests.

The results reveal a clear size-segment pattern. VN30 and VNAllshare obtain a perfect 9/9 diagnostic pass rate, indicating

that the AR(1)-GARCH-type framework is adequate for large-cap and broad-market returns. VNMidcap passes 4 out of 9 specifications, while VNSmallcap fails all specifications because of remaining autocorrelation in standardised residuals.

The fact that all models pass the squared-residual and ARCH-LM diagnostics suggests that the variance equations remove most remaining conditional heteroskedasticity. However, it does not prove complete variance-model correctness. The remaining failures for VNMidcap and VNSmallcap are mainly mean-equation failures rather than volatility-equation failures. A plausible explanation is non-synchronous trading and thinner liquidity in smaller-size segments, but this cannot be directly verified using only index-level daily closing prices.

E. In-Sample VaR Backtesting Results

Table VI reports Kupiec unconditional coverage p -values for selected in-sample one-step-ahead VaR forecasts. The table includes the GARCH(1,1)-Normal and GARCH(1,1)-Student- t baselines and the EGARCH-SkewT model selected by AIC/BIC. A p -value above 0.05 indicates that the empirical failure rate is statistically indistinguishable from the nominal VaR level.

TABLE VI
IN-SAMPLE VAR BACKTESTING: KUPIEC p -VALUES

Index	Model	$\alpha = 1\%$		$\alpha = 5\%$	
		Long	Short	Long	Short
VN30	Normal	<0.001	0.008	0.879	<0.001
VN30	Stud.- t	<0.001	<0.001	0.026	<0.001
VN30	EGARCH-SkewT	0.132	0.004	0.913	0.913
VNAllshare	Normal	<0.001	<0.001	0.778	<0.001
VNAllshare	Stud.- t	<0.001	<0.001	<0.001	<0.001
VNAllshare	EGARCH-SkewT	0.064	0.008	0.983	0.649
VNMidcap	Normal	<0.001	<0.001	0.377	<0.001
VNMidcap	Stud.- t	<0.001	<0.001	<0.001	<0.001
VNMidcap	EGARCH-SkewT	0.184	0.047	0.743	0.333
VNSmallcap	Normal	<0.001	0.002	0.070	<0.001
VNSmallcap	Stud.- t	0.007	<0.001	<0.001	<0.001
VNSmallcap	EGARCH-SkewT	0.886	0.255	0.948	0.222

Bold values indicate Kupiec pass at the 5% level. Normal and Student- t rows represent GARCH(1,1) specifications. EGARCH-SkewT is shown because it is the best in-sample specification by AIC/BIC.

The Normal specification performs poorly in the 1% tail and in most short-position cases, although it passes the Kupiec test for several 5% long-position cases. Hence, its weakness is concentrated in extreme-tail and asymmetric-tail coverage rather than in every VaR case. The symmetric Student- t GARCH model performs worse than the Normal baseline in this table: it corrects for heavy tails but fails to model the left-right asymmetry of the return distribution.

The Skewed Student- t distribution substantially improves in-sample VaR coverage. Among all SkewT specifications, APARCH-SkewT records the highest in-sample main pass rate at 81.25%, while EGARCH-SkewT and GJR-GARCH-SkewT each pass 75.00% of the in-sample tests. This confirms that skewness in the innovation distribution is statistically important for VaR modelling.

However, 1% short-position VaR remains difficult for VN30, VNAllshare, and VNMidcap even under EGARCH-SkewT. This indicates that the right tail is over-covered in those cases: the model assigns more probability to large positive returns than is empirically warranted.

F. Out-of-Sample Rolling VaR Backtesting Results

1) *Overall Model Ranking:* Table VII reports the overall out-of-sample rolling VaR performance for all nine model specifications, evaluated across 16 tests: four indices, two trading positions, and two VaR levels.

TABLE VII
OUT-OF-SAMPLE ROLLING VaR: OVERALL PERFORMANCE

Model	Dist.	Main pass	Kupiec pass	Christ. pass	DQ pass
GARCH(1,1)	SkewT	75.0%	75.0%	87.5%	93.8%
GJR-GARCH	SkewT	68.8%	75.0%	87.5%	62.5%
EGARCH	SkewT	62.5%	75.0%	87.5%	62.5%
APARCH	SkewT	62.5%	75.0%	68.8%	56.3%
GARCH(1,1)	Normal	25.0%	25.0%	25.0%	50.0%
APARCH	Stud.-t	0.0%	6.3%	12.5%	31.3%
EGARCH	Stud.-t	0.0%	6.3%	0.0%	25.0%
GARCH(1,1)	Stud.-t	0.0%	6.3%	0.0%	25.0%
GJR-GARCH	Stud.-t	0.0%	0.0%	12.5%	37.5%

Pass rates are computed out of 16 tests. Main pass is defined as joint Kupiec and Christoffersen conditional coverage pass at the 5% level.

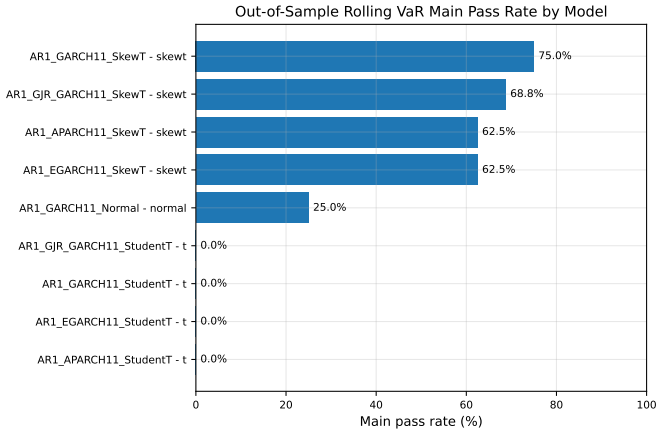


Fig. 4. Overall out-of-sample main pass rates by model. SkewT specifications dominate Normal and Student-*t* alternatives.

Figure 4 makes the main out-of-sample result visually clear. The ranking is driven primarily by the innovation distribution rather than by the variance equation. The four SkewT specifications occupy the top positions, while all symmetric Student-*t* specifications fail the main backtesting criterion in all 16 cases. This indicates that fat tails alone are not sufficient for Vietnamese equity VaR forecasting; tail asymmetry must also be modelled.

The distinction between the three pass rates is also informative. Kupiec pass rates evaluate only whether the number of violations is correct, while Christoffersen pass rates also penalise violation clustering. DQ pass rates are stricter because they test whether violations are predictable from their own

lags and the VaR forecast. The GARCH(1,1)-SkewT model performs well across all three dimensions. Its 93.8% DQ pass rate is particularly important because it indicates that the hit sequence contains little remaining dynamic structure. Hence, the model is not merely matching the unconditional violation frequency; it is also producing comparatively well-behaved dynamic VaR forecasts.

Among the SkewT specifications, the simpler GARCH(1,1)-SkewT model outperforms more complex EGARCH and APARCH specifications in out-of-sample backtesting. Its main pass rate is 75.0%, while EGARCH-SkewT and APARCH-SkewT each reach 62.5%. Economically, this means that the best model for risk forecasting is not necessarily the model with the best in-sample likelihood fit. In this sample, a parsimonious variance equation combined with a flexible skewed and fat-tailed distribution provides the strongest overall VaR performance.

2) *The 1% Long VaR Failure:* One specific result warrants attention. At the 1% long-VaR level, VN30 and VNAllshare fail the Kupiec test under several SkewT specifications in the out-of-sample window. Under EGARCH-SkewT, for example, both VN30 and VNAllshare record an empirical failure rate of 1.52% against the nominal target of 1.00%, with Kupiec *p*-values of 0.041.

This should be interpreted as downside tail under-coverage for large-cap and broad-market risk. The failure is economically meaningful because long-position VaR is the relevant measure for investors exposed to market drawdowns. However, the magnitude is moderate: the empirical failure rate exceeds the target by about 0.52 percentage points. VNSmallcap, in contrast, passes the 1% long-VaR test under EGARCH-SkewT, with a failure rate of 1.07% and a Kupiec *p*-value of 0.770.

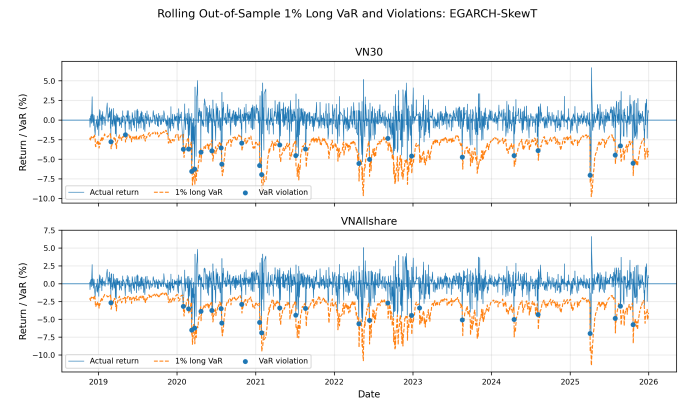


Fig. 5. Rolling out-of-sample 1% long VaR forecasts and realised returns for VN30 and VNAllshare under EGARCH-SkewT. Dots indicate VaR violations.

Figure 5 provides a direct visual interpretation of the 1% long-VaR failure. Most realised returns remain above the model-implied VaR threshold, but several negative observations fall below the threshold and are recorded as violations. Therefore, the Kupiec rejection does not mean that the model fails continuously; it means that violations occur more often than the nominal 1% frequency allows. This distinction is

important because the model remains informative for daily risk monitoring, even though it under-covers the most extreme downside tail for VN30 and VNAllshare.

G. Best Models by Index, Position, and VaR Level

Table VIII identifies the top-ranked model for each index–position–VaR level combination in the rolling out-of-sample evaluation.

TABLE VIII
BEST OUT-OF-SAMPLE MODEL BY INDEX, POSITION, AND VaR LEVEL

Index	Position	Level	Best Model	Fail rate
VN30	Long	1%	EGARCH-SkewT	1.52%
VN30	Long	5%	GJR-SkewT	5.24%
VN30	Short	1%	EGARCH-SkewT	0.73%
VN30	Short	5%	GARCH(1,1)-SkewT	5.35%
VNAllshare	Long	1%	EGARCH-SkewT	1.52%
VNAllshare	Long	5%	EGARCH-SkewT	5.24%
VNAllshare	Short	1%	APARCH-SkewT	0.90%
VNAllshare	Short	5%	GJR-SkewT	5.12%
VNMidcap	Long	1%	GARCH(1,1)-SkewT	1.41%
VNMidcap	Long	5%	GJR-SkewT	5.35%
VNMidcap	Short	1%	GARCH(1,1)-SkewT	0.73%
VNMidcap	Short	5%	GARCH(1,1)-SkewT	5.69%
VNSmallcap	Long	1%	EGARCH-SkewT	1.07%
VNSmallcap	Long	5%	EGARCH-SkewT	5.29%
VNSmallcap	Short	1%	GJR-SkewT	0.90%
VNSmallcap	Short	5%	GARCH(1,1)-SkewT	5.63%

Best model is selected by joint Kupiec–Christoffersen pass and then by the smallest absolute failure-rate error. All best models use the Skewed Student- t distribution.

No single variance specification dominates every case. EGARCH-SkewT is selected in 6 out of 16 best-model cases overall, including 4 out of 8 cases across VN30 and VNAllshare. GARCH(1,1)-SkewT is selected in 5 out of 16 cases and is especially strong for VNMidcap, where it wins 3 out of 4 cases. GJR-SkewT is selected in 4 cases, while APARCH-SkewT is selected only once.

The common element is the distributional assumption: every best model uses the Skewed Student- t distribution. This reinforces the conclusion that, for VaR forecasting in this sample, distributional specification matters more than choosing the most flexible variance equation.

H. Long versus Short VaR Performance

A consistent asymmetry between long and short VaR performance appears in the rolling out-of-sample results. Table IX compares average empirical failure rates across all SkewT models for long and short positions at the 1% VaR level.

TABLE IX
AVERAGE FAILURE RATES: LONG VS. SHORT POSITION (1% VaR, SKEWT MODELS)

Index	Long	Short	Difference
VN30	1.591%	0.648%	+0.943 pp
VNAllshare	1.633%	0.760%	+0.873 pp
VNMidcap	1.548%	0.746%	+0.802 pp
VNSmallcap	1.084%	0.887%	+0.197 pp

Averages are computed across the four SkewT variance specifications. Positive difference means the long-position failure rate exceeds the short-position failure rate.

Long failure rates exceed short failure rates for all four indices. The largest gap is observed for VN30, while the smallest gap is observed for VNSmallcap. This pattern has a clear economic interpretation: downside tail risk is more difficult to forecast than upside tail risk in Vietnamese equity returns.

The smaller long–short gap for VNSmallcap should not be interpreted as lower negative skewness. The SkewT parameter estimates show that VNSmallcap has the strongest left skewness among the four indices. Rather, the smaller gap means that the fitted SkewT VaR models produce more balanced empirical coverage between the two tails for this segment.

I. Size-Segment Risk Differences

Table X summarises average absolute VaR forecasts and average failure rates by index and VaR level. The risk rank is assigned by average absolute 1% long VaR.

TABLE X
SIZE-SEGMENT RISK SUMMARY: AVERAGE ABSOLUTE VaR AND FAILURE RATES

Index	Position	Avg. VaR (pp)		Fail rate (1% long)	Rank (1% long)
		1%	5%		
VNMidcap	Long	3.587	2.092	1.939%	1
VN30	Long	3.254	1.871	1.883%	2
VNAllshare	Long	3.232	1.827	2.002%	3
VNSmallcap	Long	3.208	1.877	1.508%	4

Averages are computed across rolling out-of-sample forecasts and all nine model specifications. Risk rank is assigned by average absolute 1%-long VaR, with the largest value ranked as the riskiest.

VNMidcap has the highest average absolute VaR at both the 1% and 5% levels, making it the riskiest segment by this metric. This result is economically meaningful because it shows that size-segment risk is not monotonic: the small-cap index is not automatically the riskiest once risk is measured using daily index-level VaR.

VN30 and VNAllshare have similar VaR levels, consistent with their high return correlation. VNSmallcap has the lowest average absolute 1% long VaR in this table, but this should not be interpreted as evidence that small-cap exposure is low-risk in general. Daily close-to-close index VaR does not capture illiquidity, bid–ask spreads, stock-level idiosyncratic risk, or

execution risk. Therefore, the size-segment comparison should be interpreted as a comparison of model-implied index-level tail risk, not total investable portfolio risk.

J. Expected Shortfall and Average Multiple of Tail Events

Table XI reports empirical Expected Shortfall (ES) and the Average Multiple of Tail Event to VaR (AM) for the best model at each index–position–level combination, computed from rolling out-of-sample forecasts.

TABLE XI
EXPECTED SHORTFALL AND AVERAGE MULTIPLE (BEST SKEWT MODELS)

Index	Pos.	Level	ES (pp)	Avg. excess	AM	Pass
VN30	Long	1%	−4.454	1.242	1.450	No
VN30	Long	5%	−3.206	1.227	1.678	Yes
VN30	Short	1%	2.706	0.366	1.162	Yes
VN30	Short	5%	2.348	0.499	1.282	Yes
VNAllshare	Long	1%	−4.508	1.281	1.484	No
VNAllshare	Long	5%	−3.201	1.279	1.724	Yes
VNAllshare	Short	1%	2.783	0.512	1.237	Yes
VNAllshare	Short	5%	2.159	0.517	1.340	Yes
VNMidcap	Long	1%	−4.335	1.197	1.429	Yes
VNMidcap	Long	5%	−3.404	1.194	1.624	Yes
VNMidcap	Short	1%	2.655	0.309	1.135	Yes
VNMidcap	Short	5%	2.280	0.456	1.259	Yes
VNSmallcap	Long	1%	−3.981	1.269	1.525	Yes
VNSmallcap	Long	5%	−3.094	0.989	1.550	Yes
VNSmallcap	Short	1%	2.295	0.243	1.143	Yes
VNSmallcap	Short	5%	1.969	0.415	1.283	Yes

ES is the average actual return on violation days. Avg. excess is the average amount by which realised tail observations exceed the VaR threshold. AM is the average multiple of realised tail loss relative to the VaR forecast. Pass indicates whether the Kupiec test passes at the 5% level.

The ES results show that long-position violations are economically more severe than short-position violations. At the 1% level, long-position ES ranges from −3.981 to −4.508 percentage points, while short-position ES ranges only from 2.295 to 2.783 percentage points. This confirms the practical importance of downside risk for long-only investors.

The AM values for 1% long VaR range from 1.429 to 1.525. Thus, conditional on a violation, realised losses exceed the VaR forecast by approximately 43–53% on average. For short positions, AM values are lower, consistent with a thinner right tail. These findings are broadly consistent with the use of ES and average tail multiples in the VaR literature, including Diamandis et al. [13].

The AM values at the 5% long level are larger than those at the 1% long level. This does not necessarily mean that the 5% VaR has worse statistical coverage. Coverage and exceedance severity measure different dimensions of tail risk: the former concerns how often violations occur, while the latter concerns how large violations are once they occur.

K. Summary of Hypothesis Testing

The empirical results allow a systematic evaluation of the four research hypotheses stated in Section I.

H1: *Vietnamese equity returns are stationary but non-normal, fat-tailed, negatively skewed, and conditionally heteroskedastic.*

Supported. ADF tests confirm stationarity for all four return series. Jarque–Bera tests reject normality. Ljung–Box tests on squared returns and ARCH-LM tests confirm conditional heteroskedasticity. The SkewT parameter estimates further show negative skewness across all model–index combinations.

H2: *Skewed Student-*t* models outperform Normal and Student-*t* models for VaR forecasting.* Strongly supported. The Skewed Student-*t* distribution dominates both in-sample and out-of-sample evaluations. In the rolling out-of-sample backtesting, all Student-*t* models have a 0% main pass rate, while SkewT models achieve main pass rates between 62.5% and 75.0%.

H3: *Asymmetric variance-equation models outperform symmetric GARCH(1,1).* Partially supported. EGARCH-SkewT has the best in-sample AIC and BIC for all four indices, confirming the value of asymmetric volatility modelling for goodness of fit. However, the simpler GARCH(1,1)-SkewT model delivers the best out-of-sample VaR backtesting performance, with a 75.0% main pass rate. Thus, asymmetric variance equations improve in-sample fit but do not necessarily improve rolling VaR forecasts.

H4: *VaR accuracy and risk levels differ systematically across size segments, VaR levels, and long/short positions.* Supported. The four indices share a strong common market component, as shown by their price co-movement, return shocks, conditional volatility spikes, and high return correlations. However, this common component does not eliminate segment-level differences in tail-risk modelling. Diagnostic pass rates differ sharply across segments: VN30 and VNAllshare pass 9/9 specifications, VNMidcap passes 4/9, and VNSmallcap passes 0/9. VNMidcap has the highest average absolute VaR. Long-position failure rates exceed short-position failure rates for all indices at the 1% level. No single model wins across all 16 index–position–level combinations, so segment-specific model selection is warranted.

V. DISCUSSION

A. Economic Interpretation

The empirical results show that Vietnamese equity returns display the main stylised facts usually observed in financial return series: non-normality, fat tails, skewness, serial dependence in selected indices, and strong conditional heteroskedasticity. These findings are economically important because they imply that market risk cannot be adequately described by a constant-variance model or by a Gaussian return distribution. In particular, the strong rejection of normality and the significant ARCH-LM results indicate that volatility is time-varying and clustered. Therefore, the use of GARCH-type models is not only a technical modelling choice but a direct response to the empirical structure of the data.

The figures added to the empirical section make this point clearer. The four indices move together in levels, returns, and conditional volatility because they are exposed to the same national market environment. This similarity is not a weakness of the research design. It is precisely the condition under which the size-segment question becomes meaningful: if the indices

share the same market factor, then any remaining differences in VaR performance, diagnostic adequacy, or tail asymmetry are more informative than simple visual differences in price paths.

The visual evidence also shows that the four indices share a strong common market component. Their normalised price paths, daily return shocks, and conditional volatility spikes tend to move together during major market-wide episodes. This co-movement is expected because all indices are drawn from the same national equity market. However, common direction of movement does not imply identical risk dynamics. The empirical results show that segment-level differences remain visible in skewness, volatility persistence, residual diagnostics, VaR failure rates, and best-model selection. Thus, the evidence supports segment-specific VaR modelling not because the four indices are independent, but because their tail-risk behaviour is not identical despite their shared market factor.

The results also show that the assumption of normally distributed innovations is too restrictive for Vietnamese equity VaR forecasting. The Normal GARCH(1,1) model performs poorly in the extreme tail and in most short-position cases. Although it passes some 5% long-position Kupiec tests in-sample, its overall performance is weak once the evaluation moves to the rolling out-of-sample setting. This confirms that a model may appear acceptable in moderate in-sample cases but still fail as a tail-risk forecasting tool. For VaR applications, the relevant question is not whether a model can describe average volatility, but whether it can forecast tail quantiles with correct violation frequency and independence.

The dominance of Skewed Student- t innovations is the most robust result of the paper. All best out-of-sample models use the Skewed Student- t distribution, and the SkewT specifications strongly outperform both Normal and symmetric Student- t alternatives. This result has a clear economic meaning. Vietnamese equity returns are not only fat-tailed; they are also asymmetric. A symmetric Student- t distribution can accommodate excess kurtosis, but it cannot fully capture the difference between the left and right tails. Since the estimated skewness parameters are negative across all indices and model specifications, downside tail risk is structurally more important than upside tail risk in this sample. This is why models that jointly allow for fat tails and skewness perform better in VaR backtesting.

The comparison between in-sample model fit and out-of-sample VaR performance is also important. EGARCH-SkewT is selected by AIC and BIC for all four indices, which shows that asymmetric volatility dynamics improve in-sample likelihood fit. However, the best out-of-sample VaR model is GARCH(1,1)-SkewT, not EGARCH-SkewT or APARCH-SkewT. This difference means that the best-fitting volatility model is not necessarily the best forecasting model. More flexible variance equations can improve in-sample fit, but their additional parameters may also increase estimation uncertainty in rolling windows. For risk forecasting, parsimony can be more valuable than maximum in-sample flexibility.

The size-segment results show that risk differs across

Vietnamese equity segments even though the indices share a common market component. VN30 and VNAllshare have the strongest residual diagnostic performance, with all nine specifications passing the diagnostic criteria. VNMidcap and VNSmallcap are harder to model, mainly because of remaining autocorrelation in standardised residuals. This suggests that the AR(1) mean equation is adequate for large-cap and broad-market returns but less adequate for smaller-size segments. The issue is not primarily remaining ARCH effects, because the squared-residual and ARCH-LM diagnostics are passed across all models. Instead, the remaining problem is concentrated in the conditional mean equation.

VNMidcap records the highest average absolute VaR, while VNSmallcap has lower average absolute index-level VaR but stronger estimated left skewness. This distinction matters. Risk is not captured by volatility or VaR magnitude alone. A segment can have lower average VaR but still exhibit worse tail asymmetry, lower liquidity, higher idiosyncratic risk, or larger trading costs. Since this study uses daily index-level closing prices, the size-segment results should be interpreted as evidence on index-level market risk, not as a full measure of investable portfolio risk.

B. Implications for Investors and Risk Managers

The first implication is that Normal GARCH should not be used as the main VaR benchmark for Vietnamese equity indices. Its weakness is most visible in extreme-tail and asymmetric-tail coverage. A risk manager relying on a Normal GARCH model would likely underestimate or misallocate tail risk, especially when evaluating the 1% VaR level or comparing long and short positions. The empirical evidence supports replacing Normal GARCH with a Skewed Student- t specification as the baseline model.

The second implication is that large-cap and broad-market risk can be modelled more reliably than mid-cap and small-cap risk using standard univariate GARCH-type models. VN30 and VNAllshare show strong diagnostic performance, while VNMidcap and VNSmallcap display remaining mean-equation dependence. For institutional investors whose exposure is concentrated in VN30-like portfolios, the fitted GARCH-type VaR framework is relatively more reliable. For mid-cap and small-cap exposures, VaR forecasts should be interpreted with greater caution, and the model should be supplemented with additional diagnostics or alternative mean specifications.

The third implication concerns long versus short risk. Long-position failure rates exceed short-position failure rates at the 1% level for all four indices. This means that downside risk is more difficult to forecast than upside risk. For long-only investors, this is the economically relevant tail: losses occur when the market falls. Therefore, the 1% long VaR results should receive more weight than short-position results when the objective is capital protection or downside risk control.

The fourth implication is that the 1% VaR level should be interpreted more cautiously than the 5% level. The 1% tail contains fewer violations, so test results are more sensitive to a small number of extreme events. In the out-of-sample

results, VN30 and VNAllshare fail the 1% long-VaR Kupiec test under several SkewT specifications, even though their overall model performance remains strong. This does not make the models useless; it shows that very extreme downside risk remains difficult to forecast during stress periods. In practice, 1% VaR should be reported together with Expected Shortfall and exceedance severity measures, not used alone.

The fifth implication is that model selection should be segment-specific. The best model differs across index, position, and VaR level. EGARCH-SkewT is frequently selected for VN30 and VNAllshare, while GARCH(1,1)-SkewT performs especially well for VNMidcap. No single variance equation dominates all 16 cases. The only stable feature across the best models is the Skewed Student- t distribution. Therefore, a practical risk system should treat SkewT innovations as the default distributional assumption, while allowing the variance equation to vary by segment and use case.

C. Methodological Implications

The results imply that VaR model evaluation should not rely only on AIC, BIC, or log-likelihood. These criteria measure in-sample statistical fit, but VaR is a forecasting problem. A model selected by AIC may fit the conditional density well in-sample but still produce inferior rolling VaR forecasts. This paper demonstrates that point clearly: EGARCH-SkewT is preferred by AIC/BIC, while GARCH(1,1)-SkewT has the best overall out-of-sample backtesting performance.

A proper VaR evaluation framework should combine several criteria. The failure rate checks whether the number of violations is close to the nominal level. The Kupiec test evaluates unconditional coverage. The Christoffersen test evaluates whether violations are independent and conditionally well specified. The Dynamic Quantile test checks whether the violation process is predictable from past information. Expected Shortfall and Average Multiple measures then describe the severity of violations after they occur. These criteria answer different questions and should not be collapsed into a single in-sample fit statistic.

The findings also show that distributional specification is more important than volatility-equation complexity in this application. Moving from Normal or symmetric Student- t innovations to Skewed Student- t innovations produces a larger improvement than moving from GARCH to EGARCH, GJR-GARCH, or APARCH. This is consistent with the economic structure of the data: the key problem is not only volatility clustering but also asymmetric tail behaviour. A complex variance equation cannot compensate for a misspecified innovation distribution.

The diagnostic results also suggest that the mean equation deserves more attention in smaller-size segments. The current AR(1) specification is sufficient for VN30 and VNAllshare, but VNMidcap and VNSmallcap retain standardised-residual autocorrelation. This means that future model designs should not treat the mean equation as a purely mechanical pre-filter. Alternative ARMA specifications, higher-order autoregressive

terms, or liquidity-adjusted return measures may be necessary for smaller segments.

D. Limitations

Several limitations should be acknowledged.

First, the analysis uses index-level returns rather than individual stock returns. Index data are useful for comparing broad size segments, but they hide firm-level heterogeneity. Small-cap risk, in particular, may be understated at the index level because index aggregation smooths idiosyncratic shocks.

Second, the common sample begins in 2014. This ensures that all four indices are compared over the same period, but it excludes earlier phases of the Vietnamese stock market. As a result, the findings may not capture volatility behaviour during older market regimes.

Third, the study uses daily close-to-close returns. Daily data are standard in VaR studies, but they do not capture intraday volatility, liquidity shortages, price-limit dynamics, or execution costs. These omitted features are especially relevant for mid-cap and small-cap exposures.

Fourth, the models are univariate. Each index is modelled separately, so the framework does not capture volatility spillovers or dynamic correlations across size segments. This is a constraint because the correlation matrix shows strong comovement among the four indices, especially between VN30 and VNAllshare and between VNMidcap and VNSmallcap.

Fifth, the models do not include macroeconomic or global risk variables. Vietnamese equity volatility may respond to exchange rates, interest rates, foreign capital flows, global equity shocks, oil prices, or global risk sentiment. Since these variables are omitted, the models are purely return-based and may react slowly to new external information.

Sixth, the rolling out-of-sample design refits the models every 20 trading days rather than every day. This design is computationally practical and reduces estimation noise, but daily refitting would be a stricter forecasting exercise. The results may differ if the models are updated more frequently.

Seventh, VaR itself has a conceptual limitation: it measures a quantile, not the full severity of losses beyond the threshold. This paper partly addresses that weakness by reporting Expected Shortfall and Average Multiple measures. Nevertheless, the core framework remains VaR-centred, and therefore it cannot fully describe the loss distribution beyond the VaR cutoff.

Finally, the results are sensitive to data construction choices, index composition, market microstructure, and liquidity conditions. The findings should therefore be interpreted as evidence for the selected indices, sample period, and modelling design, not as universal laws for all Vietnamese equity portfolios.

E. Future Research

Future research can extend the analysis in several directions.

First, the GARCH-type VaR models can be compared with Filtered Historical Simulation, historical simulation, and stochastic volatility models. This would show whether parametric GARCH models remain competitive against semi-parametric or simulation-based approaches.

Second, future work can apply CAViaR or quantile regression VaR models. These models estimate the conditional quantile directly instead of first modelling the full conditional variance and innovation distribution. They may be useful when the main objective is quantile forecasting rather than volatility modelling.

Third, multivariate volatility models such as DCC-GARCH can be used to model time-varying correlations and volatility spillovers across VN30, VNMidcap, VNSmallcap, and VNAllshare. This would be more suitable for portfolio-level risk management than separate univariate models.

Fourth, the framework can be extended to VN30 futures. Futures data would allow the analysis to connect VaR forecasts with margin requirements, hedging performance, and trading strategies. It would also make short-position risk more directly relevant.

Fifth, macroeconomic and global variables can be incorporated into the mean or variance equation. Candidate variables include VIX, USD/VND, domestic interest rates, oil prices, foreign investor trading flows, and global equity returns. These variables may improve tail-risk forecasts during externally driven stress episodes.

Sixth, the sample can be divided into subperiods such as pre-COVID, COVID, and post-COVID periods. This would test whether the dominance of SkewT models is stable across regimes or driven mainly by crisis observations.

Overall, the discussion points to one central conclusion: for Vietnamese equity size segments, accurate VaR forecasting depends more on correctly modelling asymmetric and fat-tailed innovations than on selecting the most complex volatility equation. However, smaller-size segments require additional care because their residual dynamics are not fully captured by the current AR(1)-GARCH-type framework.

VI. CONCLUSION

This paper investigates volatility and one-day-ahead Value-at-Risk forecasting across Vietnamese equity size segments using GARCH-type models. The analysis focuses on four indices: VN30, VNMidcap, VNSmallcap, and VNAllshare. These indices represent large-cap, mid-cap, small-cap, and broad-market exposures within the Vietnamese equity market. The empirical framework estimates AR(1)-GARCH-type models under Normal, Student- t , and Skewed Student- t innovation distributions, and evaluates VaR forecasts for both long and short trading positions at the 1% and 5% levels. The purpose is to determine whether volatility and tail-risk forecasting performance differs across size segments and whether flexible conditional distributions improve VaR accuracy.

The preliminary evidence confirms that Vietnamese equity returns display the main stylised facts of financial return series. The return series are stationary, but they are clearly non-normal, fat-tailed, negatively skewed, and conditionally heteroskedastic. Ljung-Box tests on squared returns and ARCH-LM tests provide strong evidence of volatility clustering across all indices. These findings justify the use of GARCH-type volatility models. The estimated Skewed Student- t parameters further show that downside tail asymmetry is present in all

four size segments, with stronger left skewness in smaller-size indices.

The model comparison produces two main findings. In-sample, EGARCH with Skewed Student- t innovations provides the best fit for all four indices according to AIC and BIC. This shows that asymmetric volatility responses and skewed fat-tailed innovations are important for describing the conditional distribution of Vietnamese equity returns. Out-of-sample, however, the ranking changes. GARCH(1,1) with Skewed Student- t innovations delivers the strongest rolling VaR backtesting performance, with the highest overall main pass rate across indices, positions, and VaR levels. This result demonstrates that the best-fitting model is not necessarily the best forecasting model. For VaR forecasting, distributional specification matters more than volatility-equation complexity.

The long-short comparison also shows that downside risk is more difficult to forecast than upside risk. At the 1% level, long-position failure rates exceed short-position failure rates for all four indices. The Expected Shortfall and Average Multiple results further indicate that long-position violations are economically more severe than short-position violations. These findings are important for long-only investors and risk managers because the main practical risk in equity portfolios is large negative returns. The size-segment results show that VNMidcap has the highest average absolute VaR, while VNSmallcap exhibits the strongest left skewness and the weakest residual diagnostic performance. Therefore, size-segment risk differs not only in magnitude but also in distributional shape and model adequacy.

The main contribution of this study is to extend the long-short VaR framework to Vietnamese equity size segments. Instead of comparing broad equity indices across countries, the paper compares risk dynamics within one emerging market across large-cap, mid-cap, small-cap, and broad-market exposures. The evidence shows that a single market-wide VaR specification should not be imposed mechanically across all size segments. The results also show that Skewed Student- t innovations should be treated as the baseline distributional choice for Vietnamese equity VaR forecasting. More complex variance equations can improve in-sample fit, but they do not necessarily improve rolling out-of-sample VaR performance.

Several limitations remain. The analysis uses daily index-level data and therefore does not capture intraday volatility, stock-level heterogeneity, liquidity costs, or execution risk. The models are univariate, so they do not capture volatility spillovers or time-varying correlations across size segments. The framework also omits macroeconomic and global risk factors that may affect Vietnamese equity volatility. Future research could extend the analysis using multivariate GARCH models, Filtered Historical Simulation, CAViaR or quantile regression VaR, VN30 futures data, and subperiod analysis around stress episodes such as COVID-19 and the 2022 market sell-off.

REFERENCES

- [1] R. F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation," *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.
- [2] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [3] D. B. Nelson, "Conditional heteroskedasticity in asset returns: A new approach," *Econometrica*, vol. 59, no. 2, pp. 347–370, 1991.
- [4] L. R. Glosten, R. Jagannathan, and D. E. Runkle, "On the relation between the expected value and the volatility of the nominal excess return on stocks," *Journal of Finance*, vol. 48, no. 5, pp. 1779–1801, 1993.
- [5] Z. Ding, C. W. J. Granger, and R. F. Engle, "A long memory property of stock market returns and a new model," *Journal of Empirical Finance*, vol. 1, no. 1, pp. 83–106, 1993.
- [6] P. Jorion, *Value at Risk: The New Benchmark for Managing Financial Risk*, 3rd ed. New York, NY, USA: McGraw-Hill, 2007.
- [7] C. Alexander, *Market Risk Analysis: Value-at-Risk Models*. Chichester, U.K.: Wiley, 2008.
- [8] P. H. Kupiec, "Techniques for verifying the accuracy of risk measurement models," *Journal of Derivatives*, vol. 3, no. 2, pp. 73–84, 1995.
- [9] P. F. Christoffersen, "Evaluating interval forecasts," *International Economic Review*, vol. 39, no. 4, pp. 841–862, 1998.
- [10] R. F. Engle and S. Manganelli, "CAViaR: Conditional autoregressive Value at Risk by regression quantiles," *Journal of Business & Economic Statistics*, vol. 22, no. 4, pp. 367–381, 2004.
- [11] B. E. Hansen, "Autoregressive conditional density estimation," *International Economic Review*, vol. 35, no. 3, pp. 705–730, 1994.
- [12] P. Giot and S. Laurent, "Value-at-Risk for long and short trading positions," *Journal of Applied Econometrics*, vol. 18, no. 6, pp. 641–663, 2003.
- [13] P. F. Diamandis, A. A. Drakos, G. P. Kouretas, and L. Zarangas, "Value-at-risk for long and short trading positions: Evidence from developed and emerging equity markets," *International Review of Financial Analysis*, vol. 20, no. 3, pp. 165–176, 2011.
- [14] P. Bui Quang, T. Klein, N. H. Nguyen, and T. Walther, "Value-at-Risk for South-East Asian stock markets: Stochastic volatility vs. GARCH," *Journal of Risk and Financial Management*, vol. 11, no. 2, article 18, 2018.
- [15] N. Vo and R. Slepaczuk, "Applying hybrid ARIMA-SGARCH in algorithmic investment strategies on S&P500 index," *Entropy*, vol. 24, no. 2, article 158, 2022.
- [16] T. Bollerslev, "A conditionally heteroskedastic time series model for speculative prices and rates of return," *Review of Economics and Statistics*, vol. 69, no. 3, pp. 542–547, 1987.
- [17] C. Fernandez and M. F. J. Steel, "On Bayesian modeling of fat tails and skewness," *Journal of the American Statistical Association*, vol. 93, no. 441, pp. 359–371, 1998.
- [18] P. Lambert and S. Laurent, "Modelling financial time series using GARCH-type models and a skewed density," Université de Liège, mimeo, 2001.
- [19] D. Hendricks, "Evaluation of value-at-risk models using historical data," *Federal Reserve Bank of New York Economic Policy Review*, Apr. 1996.
- [20] Bank for International Settlements, "Amendment to the capital accord to incorporate market risks," Basel Committee on Banking Supervision, Basel, Switzerland, Jan. 1996.
- [21] M. K. P. So and P. L. H. Yu, "Empirical analysis of GARCH models in Value at Risk estimation," *Journal of International Financial Markets, Institutions and Money*, vol. 16, no. 2, pp. 180–197, 2006.
- [22] D. G. McMillan and A. E. H. Speight, "Value-at-Risk in emerging equity markets: Comparative evidence for symmetric, asymmetric and long-memory GARCH models," *International Review of Financial Analysis*, vol. 16, no. 3, pp. 208–223, 2007.
- [23] P. R. Hansen and A. Lunde, "A forecast comparison of volatility models: Does anything beat a GARCH(1,1)?" *Journal of Applied Econometrics*, vol. 20, no. 7, pp. 873–889, 2005.